

# Rodrigo Roque-Hernandez

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Health Data Science & Digital Product | Applied ML | Human-Centered Design | Clinical Analytics

## EDUCATION

### University of North Carolina at Chapel Hill

*Bachelor of Science in Information Science & Data Science*

*Awards & Involvement:*

Chapel Hill, NC

*August 2023 – May 2027*

*HSF Scholarship | CAMpFIRE Scholarship | SHPE*

## EXPERIENCE

### UX & Web Design Intern

*SAS Institute – Health & Life Sciences Division, Figma, React, User Research*

May 2025 – Aug 2025

*Cary, NC*

- Designed and built frontend interfaces for data-driven health analytics products at a leading enterprise software company
- Conducted user research to surface pain points and inform iterative design decisions for complex data visualization tools
- Collaborated across product, engineering, and design teams to deliver user-centered solutions in an agile environment

### ecDNA Research Assistant

*Brunk Cytogenomics Lab, AI@UNC – Python, PyTorch, OpenCV, pandas*

Oct. 2024 – May 2025

*Chapel Hill, NC*

- Developed deep learning computer vision pipelines (U-Net CNN) to detect and localize extra-chromosomal DNA in FISH microscopy images, improving analysis efficiency by 25%
- Built automated preprocessing workflows including normalization, augmentation, and coordinate mapping to scale biomedical imaging datasets
- Reduced manual cancer imaging review time from over a week to hours through model automation

### Director of Web Development / Technical Lead

*NAHJ at UNC & Coulture Magazine – React, Firebase, Figma, JavaScript*

Aug. 2025 – Present

*Chapel Hill, NC*

- Led cross-functional teams of developers and designers to deliver production platforms for two student organizations
- Integrated analytics tracking and UX optimization, increasing user engagement and session duration by 23%

## PROJECTS

### Alzheimer's Classification from RNASeq Data | *Python, scikit-learn, R – DATA 510*

- Reproduced peer-reviewed ML pipeline to classify Alzheimer's vs. control subjects from high-dimensional gene expression data, applying FDR correction, differential expression analysis, and logistic regression in matrix form
- Compared model performance across top 50, 100, and 1000 gene feature subsets; assessed batch effects and proposed experimental correction strategies

### Diabetes & Obesity Risk Modeling (NHANES) | *Python, R, scikit-learn – DATA 510*

- Modeled association between obesity and diabetes using NHANES 2013–2014 data; applied odds ratios, stratified confounding analysis, ANOVA, and logistic regression, reported in clinical publication format
- Selected appropriate statistical measures based on cross-sectional study design, interpreting BMI coefficients and confidence intervals for clinical relevance

## LEADERSHIP

### Executive Director → Director of External Marketing → Marketing Chair

*SolHacks – UNC's First Latinx-Oriented Hackathon, SHPE*

Jan. 2025 – Present

*Chapel Hill, NC*

- Owned end-to-end event strategy across marketing, partnerships, web, and operations for UNC's first Latinx-oriented hackathon

## TECHNICAL SKILLS

**Languages & Data:** Python, R, SQL, JavaScript, HTML/CSS | pandas, NumPy, scikit-learn, PyTorch, OpenCV, matplotlib, seaborn

**Design & Tools:** Figma, UX Prototyping, User Research, React, Firebase, Git, Unix, Jupyter Notebooks, Google Colab

**Domains:** Healthcare analytics, clinical predictive modeling, biomedical imaging, genomics, digital health products